

Nobody Moves:
The Impact of Rideshare Apps on Transportation Infrastructure, Social Hierarchies, and
Labor Rights in Bengaluru, India

Deepika Kannan

URBS 4000 Senior Seminar

December 10, 2025

Table of Contents

Abstract	2
Introduction	3
Literature Review	5
Methodology	14
Data	16
Discussion and Analysis	24
Conclusion and Future Research	28
Bibliography	30

Abstract

Bengaluru is one of the most congested cities in the world. There has been a recent surge in rideshare services utilized by affluent residents. This paper will analyze how rideshare platforms affect Bengaluru's transportation infrastructure, mobility inequalities, and increasing platform capitalism.

Utilizing semi-structured interviews conducted in-person in August 2025 with Bengaluru drivers, residents, platform owners, and experts, this paper analyzes over forty perspectives to analyze impacts of rideshare platforms. The interviews were conducted in auto-rickshaws, rideshare cars, and public buses, metros, and parks for thirty minutes to an hour, in English, Kannada, and Tamil.

The results indicated that rideshare platforms have become a daily solution for affluent residents, making them less likely to use public transit. They also indicated that they create a wider gap in transportation experiences based on class. Additionally, rideshare apps have a monopoly on private rides, which creates driver labor precarity.

This paper concludes that rideshare platforms are ultimately a setback for sustainable transportation and social equality in Bengaluru. As Bengaluru implements the new rapid metro system, it is important to consider how rideshare platforms may affect ridership.

Introduction

When I think of Bengaluru, I think of traffic. My family in India lives in Bengaluru, and during summer visits, I always noticed how long it took to travel a seemingly short distance—the ten miles that took us fifteen minutes in our American suburb took nearly 1 hour in Bengaluru. I watched as auto rickshaws and motorcycles tried to weave through from the privilege of our private, air-conditioned car, but ultimately none of us were moving. Peddlers would take advantage of the chaos to walk directly into the congested highways, tapping on car windows to sell their wares. Separate, yet united in congestion, transportation in Bengaluru struck me as an example of inequality and inefficiency.

Bengaluru experienced an IT boom in the 1990s, earning it the title the “Silicon Valley of India,” through its service of Western markets with software services and offshore development. While the boom has created jobs and drawn talent from around the country, it has also produced insular economic enclaves, where the benefits of technology growth are only enjoyed by affluent, English-speaking professionals¹. This spawned a middle class capable of affording private, low-capacity transportation², which in turn contributed to Bengaluru’s recent ranking as the second-most congested city in the world.

To create more sustainable, long-term transportation for the growing population, Bengaluru’s rapid transit system, Namma Metro, was launched in 2011, and continues to add new stations and service to this day. Although used by over one million people each day, the system struggles with delays, overcrowding, and coverage gaps³. Aside from the metro, the city

¹ AnnaLee Saxenian, *Bangalore: The Silicon Valley of Asia?* (Stanford: Stanford Center for International Development, 2001).

² World Bank, *Urban Transport in Chennai and Bangalore* (Washington, DC: World Bank, 2004).

³ Suchith Kidiyoor, “Bengaluru Metro Sets New Record: Namma Metro Hits 10.5 Lakh Daily Ridership; Yellow Line’s First Day Sees 83,000 Commuters,” *Times of India*, August 13, 2025, <https://timesofindia.indiatimes.com/city/bengaluru/yellow-line-helps-namma-metro-ridership-cross-10-lakh-mark-in-bengaluru/articleshow/123265415.cms>

utilizes Bengaluru Metropolitan Transportation Corporation buses, auto-rickshaws, personal cars, and two-wheelers⁴.

Auto-rickshaws, the bright green and yellow, three-wheeled, open-air vehicles are a popular alternative to owning a personal vehicle or riding the bus. Families of five pile into them for daily commutes to school and work. Until ten years ago, they were hailed on the street, and residents would walk to busy streets to hail an auto by holding out their hands and shouting.

Ten years ago, this process began to be replaced through rideshare platforms. First, Uber entered Indian markets, and similar aggregators followed, such as Ola, Rapido, and Namma Yatri. They started off as a niche alternative only used by the most digitally savvy, but soon became part of the mainstream commuting system.

In Bengaluru, the actively expanding Namma Metro depends on a commuter shift towards public transit. Understanding how rideshare apps may affect this is critical to implement mass public transit. Bengaluru is also experiencing a shift with platform capitalism, with food delivery and rideshare platforms replacing formerly social interactions. It is important to understand how these may affect driver livelihoods in Bengaluru. Overall, the role of rideshare apps in Bengaluru's urban context must be studied, analyzing its effect on transportation infrastructure, social hierarchies, and labor experiences.

⁴Asian Transport Observatory, *Bengaluru Urban Transport: State of Play; Insights from the Asian Transport Observatory (January 2025)*,
https://asiantransportobservatory.org/documents/337/Bangalore_urban_state_of_play.pdf

Literature Review

Existing literature analyzes the history of Bengaluru's transportation infrastructure, its future mass transit goals, the social inequalities of mobility, and platform work inequalities. These topics are relevant to understanding how rideshare platforms fit into the greater picture of Bengaluru, in terms of infrastructure, social hierarchies, and labor.

Context of Transportation in Bengaluru

For the use of rideshare platforms to become as prevalent as it is currently, the existing transportation landscape must have been primed for it to thrive. Like many developing cities, the early stages of transportation in Bengaluru were defined during British rule. In 1809, the British created the Bangalore Cantonment, with two separate regions: the "White Town" and "Black Town."⁵ The Cantonment had 'wide, tree-lined roads, orderly street grids, and open parade grounds,' while the previously inhabited Pete had 'organic street patterns, narrow lanes, and bustling markets.'⁶ The British enclave enjoyed the first centrally planned infrastructure in Bengaluru, with piped water, drainage, lighting, and even public transportation, of 'horse-drawn trams, and later buses.'⁷

These historical hierarchies created lasting infrastructural imbalances. Neighborhoods in the former Cantonment still have wider roads, while older cores remain crowded. Scholars state the post-colonial Indian cities inherited 'high-density urban areas, absence of proper control on land-use, and poor public transport.'⁸

⁵ Kakasaheb Laxman Gasti, "Colonial Urbanization: The Making of Bangalore Under British Rule," *International Journal of Creative Research Thoughts* 6, no. 1 (2018), <https://ijcrt.org/papers/IJCRT1135992.pdf>

⁶ Gasti, "Colonial Urbanization."

⁷ Gasti, "Colonial Urbanization."

⁸ Ashish Verma, S. Sreenivasulu, and N. Dash, "Achieving Sustainable Transportation System for Indian Cities – Problems and Issues," *Current Science* 100, no. 9 (2011): 1328–39, <http://www.jstor.org/stable/24076598>

Bengaluru's bus system, although extensive, is not a rapid transit system. It cannot keep pace with rising travel demands, and the journey speeds in major Indian cities are now around 17 to 26 kilometers per hour, with projections of single digit speeds by 2031⁹. Meanwhile, Namma Metro, the city's rapid metro system, has been expanding for the past fifteen years, but the network coverage remains limited, especially for first- and last-mile trips¹⁰. The conditions of unreliable transportation and a growing urban population has created a space for rideshare platforms to emerge.

The first rideshare platform to enter India was Uber in 2013, launching as a pilot program in Bengaluru. A *New York Times* article found that Uber was motivated to enter India since it presented an opportunity to 'leapfrog Western-style car-ownership culture and move directly into a society where people don't buy cars, they hail Ubers'¹¹. Uber stated that Indian cities were 'clogged with traffic, potholes, and pedestrians,' and living with public transit systems where 'buses [are] so crowded that passengers hang out the door'¹². Their goal was to create a solution for middle-class riders seeking relief from congestion—making it 'cheaper and safer to get around, especially for women'¹³.

Bengaluru's transportation system today finds itself shaped by a legacy of segregated colonial infrastructure, an underdeveloped patchwork of transit solutions, and a landscape of aggressive rideshare platforms promising a band-aid solution to its many troubles. This final point does not find much elaboration in the literature; existing research does not analyze the precise effects of rideshare platforms in Bengaluru.

⁹ Verma, Sreenivasulu, and Dash, "Achieving Sustainable Transportation System."

¹⁰ Verma, Sreenivasulu, and Dash, "Achieving Sustainable Transportation System."

¹¹ Farhad Manjoo, "Uber Wants to Rule the World. First It Must Conquer India," *New York Times*, April 14, 2017, <https://www.nytimes.com/2017/04/14/technology/uber-india.html>

¹² Manjoo, "Uber Wants to Rule the World."

¹³ Manjoo, "Uber Wants to Rule the World."

Role of Rideshare Platforms in Existing Infrastructure

Transit-oriented development is a popular and promising transportation planning trend used in many developing cities. This approach integrates land use with transportation planning by concentrating dense, mixed-use development around high capacity transit¹⁴. For example, metro stations would only be located in spaces with housing, grocery stores, and job opportunities within walking distance, and convenient foot and bike paths¹⁵. Given the severe traffic congestion and high population density, in 2019, Bengaluru developed a comprehensive draft plan for Transit-Oriented Development.

Coordination of transportation and land development is critical to reduce the auto dependency that many developing cities face¹⁶. Having integrated transportation and livelihoods are vital for the ‘neediest members of the world’s poorest countries’¹⁷. Cities located in developing countries, such as Mexico City and Santiago, have not integrated transit with land use, inadvertently creating sprawl and pushing poor residents to the periphery¹⁸. Meanwhile, when city planners proactively invest in walkable, high-density transit corridors with supportive zoning, residents of all economic backgrounds gain access to quality public transit, as seen in cities like Bogotá and Guangzhou¹⁹.

Bengaluru’s draft TOD proposal in 2019 aimed to reduce the “rapid growth of private vehicles, inadequate road networks, long commuting time[s], and traffic congestion” by encouraging public transit²⁰. The plan also noted the importance of TOD in increasing metro ridership and reducing commuting burdens²¹. However, there is a limited amount of research on

¹⁴ Mehra and Vardhan, *Analysis of TOD in Bengaluru Suburban Railway*

¹⁵ Mehra and Vardhan, *Analysis of TOD in Bengaluru Suburban Railway*

¹⁶ Cervero, *Urban Transport and Land Use in Developing Countries*

¹⁷ Cervero, *Urban Transport and Land Use in Developing Countries*

¹⁸ Cervero, *Urban Transport and Land Use in Developing Countries*

¹⁹ Cervero, *Urban Transport and Land Use in Developing Countries*

²⁰ *Draft Bengaluru Transit Oriented Development Policy 2019*

²¹ Mehra and Vardhan, *Analysis of TOD in Bengaluru Suburban Railway*

the impact of Bengaluru TOD plans. Bengaluru urban planning leader Dr. Anjali Karol Mohan critiques the TOD Draft Proposal in a news article, stating that it does not align with Metro Rail Policy and applies only to the Bengaluru urban district rather than the entire metropolitan region²². The TOD plan proposes adding it as a chapter in the Bengaluru Master Plan, rather than being a city-wide strategy. The final Bengaluru 2031 Master Plan does not explicitly mention TOD anywhere, but discusses improving accessibility to public transit by having guaranteed non-motorized, short last-mile commutes from stations²³. It does not mention behavioral change, or how informal modes of transportation such as auto-rickshaws may poach commuters from the Metro. TOD scholars identify that a strong modal shift must be made towards public transit in order for TOD to be implemented²⁴. Rideshare platforms may encourage the opposite of TOD policies, but there is no published research on how they could affect such development goals.

Aside from transit development frameworks, existing literature analyzes the impact of rideshare platforms on vehicle ownership and traffic patterns in India and other developing countries. One study found that the arrival of rideshare platforms has reduced personal vehicle ownership in India, as it has provided a cheaper and lower-commitment alternative to owning a car²⁵. Rideshare platforms replacing vehicle ownership may seem like a victory; however, rideshare apps also replace ‘nondriving modes such as mass transit, biking, and walking’ with new vehicle trips²⁶. A study in Boston showed that 42% of ride-hailing passengers would have taken public transit and 12% would have walked or biked if the service were unavailable²⁷. In the past, strikes involving Uber and Ola drivers have resulted in notably reduced traffic congestion in

²² Mohan, *Bengaluru’s Draft Transit Policy is Unlikely to Make the City more Livable*

²³ *Revised Master Plan for Bengaluru 2031: Planning District Report*

²⁴ Knowles, *Re-evaluating Transit-Oriented Development*

²⁵ Wadud and Namala, *The effects of ride-sourcing vehicles on vehicle ownership in large Indian cities*

²⁶ Behroozi, *Understanding the Impact of Ridesharing Services on Traffic Congestion*

²⁷ Behroozi, *Understanding the Impact of Ridesharing Services on Traffic Congestion*

major Indian cities such as Mumbai, New Delhi, and Bengaluru,²⁸ suggesting the outsize influence of rideshare platforms in traffic volumes. The New Delhi metro system had a record number of riders during the shutdown of ride-hailing platforms²⁹. High-capacity transit trips are replaced with low-occupancy vehicles through rideshare platforms, which increases traffic congestion. They ultimately increase Vehicle Miles Traveled (VMT), which greatly increases the congestion in Bengaluru.

Scholars describe such informal transit, primarily used in developing cities, as laissez-faire and entrepreneurial. This culture of unlicensed auto-rickshaws exists from Indian cities to Jakarta, Nigeria, and Bangkok. Critics argue that such private carriers should be “heavily regulated, if not banned outright”³⁰. The operating practice encourages drivers to cut each other off and load in passengers by stopping on busy streets. The “presence of private carriers alongside public buses and rail systems set in motion competitive pressures on formal operators,” with an example in Brazil where the quality of bus services improved to keep competition with private, affordable, informal van services³¹. However, because these ‘informal carriers’ offer “time savings, air-conditioned rides, and guaranteed seating,” they will always win the competition with public transit. The graph in Figure 1 shows that only 0.3% of people prefer the informal mode of auto-rickshaws as their only mode of transportation, illustrating that perhaps Indian residents would take a metro, should it become available³².

²⁸ Behroozi, *Understanding the Impact of Ridesharing Services on Traffic Congestion*

²⁹ Agarwal, Mani, and Telang, *Ride-hailing services on congestion: Evidence from Indian Cities*

³⁰ Cervero, *Informal Transit: Learning from the Developing World*

³¹ Cervero, *Informal Transit: Learning from the Developing World*

³² Koan Advisory Group, *Preferred Mode of Transit by Indian Commuters*, accessed December 14 2025, via Statista

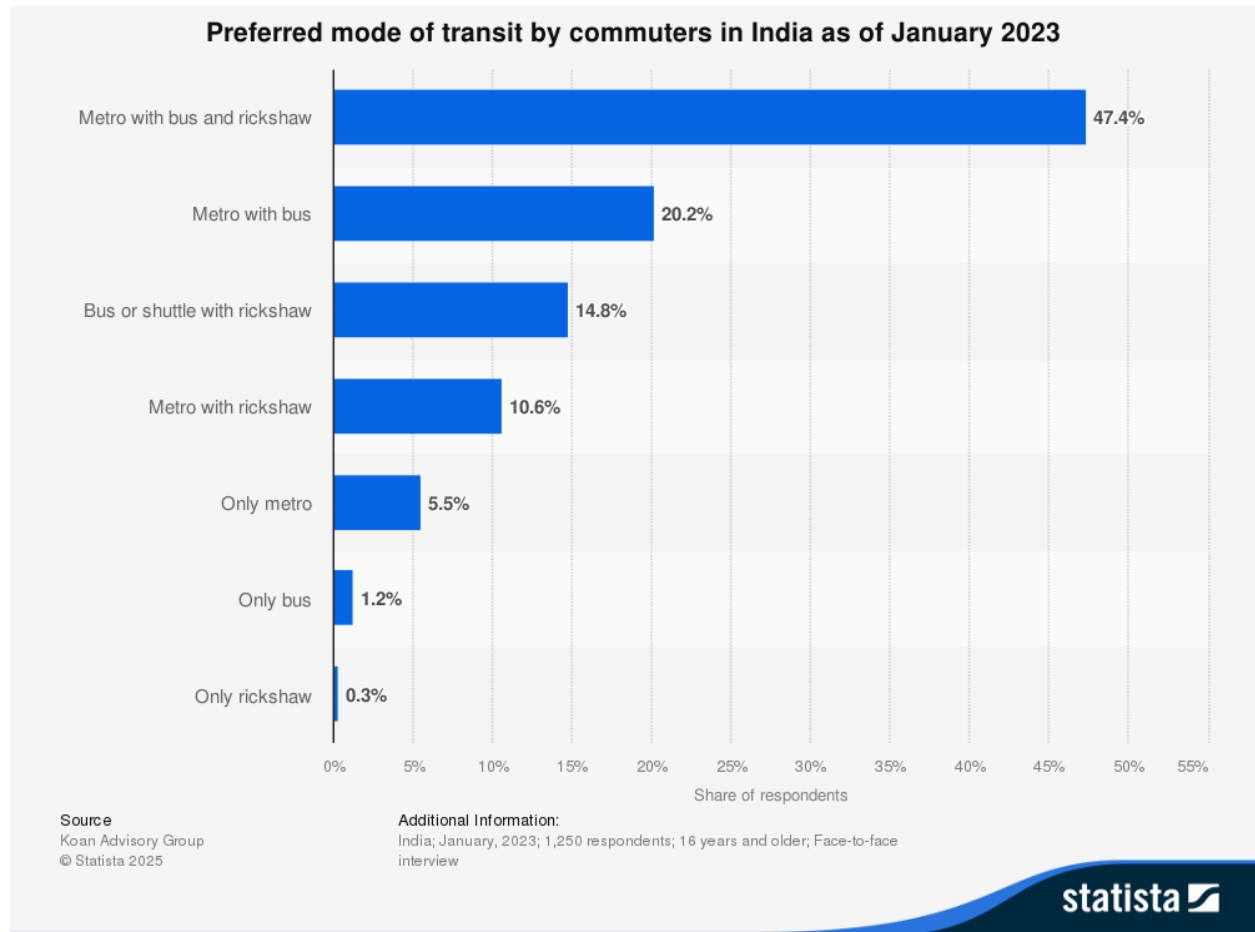


Figure 1. *Preferred mode of transit by Indian commuters.* Source: Koan Advisory Group; data accessed via Statista

From the existing literature, we can understand that Bengaluru's current city-wide transportation goals do not explicitly align with TOD principles, nor do they mention the impact of informal transportation services on Metro implementations. We can also understand that ride-hailing platforms and informal transit have negative effects on congestion and public transit usage, yet there is no study that is critical of how they will affect Bengaluru's infrastructure goals.

Rideshare Platforms and Social Hierarchies

In Bengaluru, deciding one's mode of transportation is not a neutral decision—it is a reflection of class and mobility. Critics of the social stratification of mobility access state that Bengaluru's car-centric planning efforts “privilege[s] elite users and systematically displace[s] inconvenience onto marginalized populations”³³. Poor and working-class residents bear burdens of delays and risk, as the road is “configured as a space for the smooth passage of automobility, while frictions are absorbed disproportionately by those outside the car”³⁴. In *The Social Life of a Bus*, Govind Gopakumar explains the suffering in bus travel faced by low-income residents: frequent cancellations, sweltering conditions during summer months, and routine breakdowns³⁵. Some buses with air conditioning are provided for higher fares and enjoy higher frequencies, leaving those who ride the less costly bus to suffer most. Bengaluru's revenue-driven bus routing system leaves many peripheral neighborhoods with only ‘one subsidized bus and irregular service,’ forcing vulnerable transit riders like women to spend hours waiting or to walk long distances in the dark³⁶. Women reported that reaching work late resulted in wage penalties, reflecting the direct economic impacts of transit scarcity³⁷. Inadequate transit immobilizes entire low-income communities. After slum relocations in Bengaluru, low-income women were forced into schedules that required waking at 3 a.m to access their transit routes³⁸. Mobility access is not just shaped by geographies and fares - interpersonal conflicts also affect lower-income communities' access to transit. Kalyanaraman, for instance, finds that some drivers and conductors have strained relations with people of certain neighborhoods, leading them to refuse

³³ Gopakumar, *Installing Automobility*

³⁴ Gopakumar, *Installing Automobility*

³⁵ Gopakumar, *The Social Life of a Bus*

³⁶ Kalyanaraman, ‘*When the Buses Stop, You Stop*’: *Women Commuters in Bengaluru*

³⁷ Kalyanaraman, ‘*When the Buses Stop, You Stop*’: *Women Commuters in Bengaluru*

³⁸ Kalyanaraman, ‘*When the Buses Stop, You Stop*’: *Women Commuters in Bengaluru*

to drive to some areas³⁹. Disabled women have few accessible public transit options, and are forced to use unsustainable modes of transportation such as large, low-occupancy vehicles, creating another mobility penalty⁴⁰.

Rideshare platforms enter this landscape of class-based mobility as amplifiers of existing inequalities. While there is not a study about Bengaluru social hierarchies with regard to rideshare platforms, existing scholarship suggests that car-centered reproduces the segregations that Gopakumar identifies. Rideshare platforms require smartphones, data plans, digital literacy, and the ability to pay surge prices: they are a mode only for the middle class. In a city where mobility reproduces social hierarchy, rideshare creates a new tier of premium mobility. It offers safety, speed, and predictability to those who can afford it. The last-mile gaps of Bengaluru's bus systems sometimes pushes low-income women into auto-rickshaws they cannot afford⁴¹. It deepens the existing social hierarchies where the wealthy gain more mobility choices while the poor face rising costs, longer commutes, and exclusion⁴². Rideshare platforms are not a neutral substitute for public transit: they reinforce a class and gender-based mobility hierarchy where some residents can smoothly mobilize across the city while others are financially punished for needing to travel. The graph in Figure 2 showcases the intended demographic of rideshare platforms: they cater mostly to middle-aged men and women, with more male users⁴³.

³⁹ Kalyanaraman, *'When the Buses Stop, You Stop': Women Commuters in Bengaluru*

⁴⁰ Verma and Jain, *Transportation and Quality of Life of Physically Challenged Women in Bengaluru*

⁴¹ Kalyanaraman, *'When the Buses Stop, You Stop': Women Commuters in Bengaluru*

⁴² Gopakumar, *Installing Automobility*

⁴³ Oxford Economics; Uber, *Distribution of Ride-hailing Users in Bengaluru*, accessed December 14 2025, via Statista

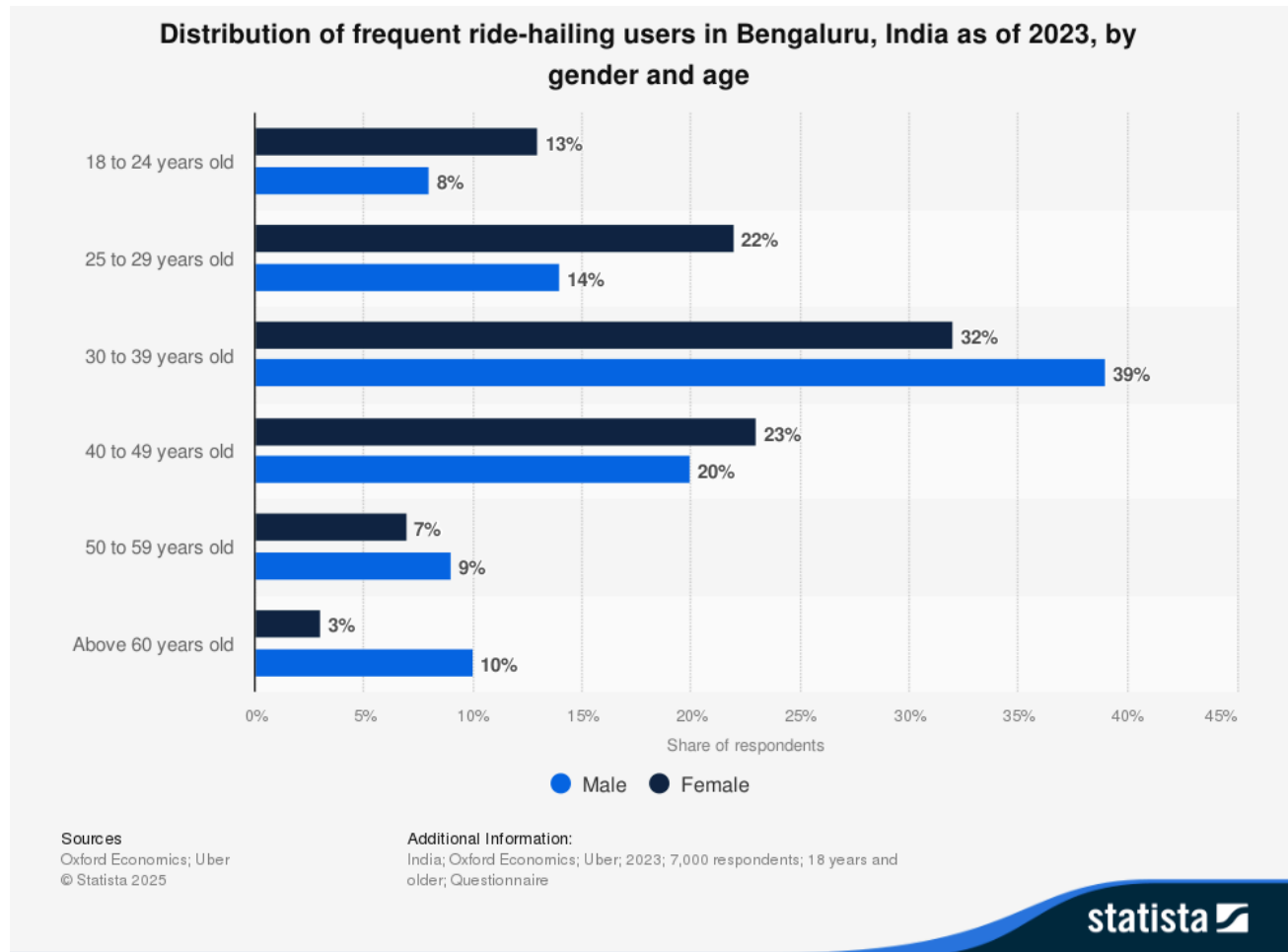


Figure 2. *Distribution of Ride-hailing Users in Bengaluru*. Source: Oxford Economics; Uber; data accessed via Statista

Rideshare Platforms and Labor Precarity

Previous investigations consistently shown that rideshare platforms are built on labor models with extreme vulnerability and precarity⁴⁴. Workers expect autonomy, flexibility, and high earnings, but studies show that ‘platforms extract labor by obscuring the true costs of driving, [and] keeping drivers in continuous debt’⁴⁵. Studies conducted on driver earnings show that they are unstable and far below what platforms initially promise. A study on auto-rickshaw

⁴⁴ Indian Federation of App-based Transport Workers, *Submission of Suggestions*

⁴⁵ Elias and Gomez, *Digital Platforms as Institutional Actors in Global South*

rideshare drivers documents that drivers experience ‘declining incentives, rising fuel costs, and falling net incomes,’ making drivers work longer hours to break even⁴⁶. Many ride-hailing drivers saw earnings decline after incentive cuts in 2017, as platforms strategically used bonuses to expand their driver base, and withdrew them once a sufficient fleet had been created. This is described by unions as a ‘race to the bottom in fares, with workers bearing the burden of price reductions’⁴⁷.

Many drivers are in debt from purchasing vehicles. Companies encourage drivers to acquire vehicles by offering them high-interest loans. The Indian Federation of App-based Transport Workers (IFAT) found that platform-affiliated vehicle loans created long-term economic dependency, with drivers unable to exit even when earnings fell⁴⁸. The loans make drivers forced to maintain a job with declining wages.

Additionally, algorithmic control is a large issue that many driver unions in India protest against⁴⁹. Previously, workers could interact with human managers, but now they are entirely managed by opaque algorithms. The algorithmic systems affect wages, trip allocations, and driver deactivation. Drivers get automatically deactivated from platforms, sometimes by an accidental low rating from passengers. The workers have little recourse when penalized, as they have nobody to report to when they get deactivated⁵⁰. Drivers feel permanently at risk due to these unclear rules⁵¹.

⁴⁶ Gutta and Bhuyan, *Impact of Using Ride Hailing Apps on Social Connectedness between Auto Drivers*

⁴⁷ Indian Federation of App-based Transport Workers, *Submission of Suggestions*

⁴⁸ Indian Federation of App-based Transport Workers, *Submission of Suggestions*

⁴⁹ Indian Federation of App-based Transport Workers, *Submission of Suggestions*

⁵⁰ Elias and Gomez, *Digital Platforms as Institutional Actors in Global South*

⁵¹ Gutta and Bhuyan, *Impact of Using Ride Hailing Apps on Social Connectedness between Auto Drivers*

Platform labor has rewired the social relations in transportation. Prior to ride-hailing apps, auto-rickshaw and taxi drivers maintained regular customers and reciprocal relationships. These social relations have been replaced with platforms that create ‘transactional, ratings-driven interactions that leave drivers feeling devalued’⁵². The ratings are arbitrary and punitive, and directly affect income and job security. Drivers who previously had long-term employer relationships now find themselves isolated and lacking a voice⁵³. Social interactions such as unions and employer-employee relations are irrelevant with platform-based work.

Platform drivers face occupational risks. Since drivers are classified as independent contractors, platforms do not assume responsibility for safety equipment or health protections⁵⁴. There is no rest requirement, insurance coverage, or health safeguards, and workers must manage illness and financial shocks independently⁵⁵.

IFAT has previously submitted requests to the Karnataka Labor Department, calling for algorithmic transparency, dispute resolutions, and social security contributions from platforms. In Bengaluru, app-based contract work extends beyond ride-hailing: food delivery and at-home services are also conducted through apps, furthering the ‘platform-driven restructuring of urban labor markets’⁵⁶.

Existing literature does not analyze how ride-hailing platforms are contributing to greater issues in Bengaluru of platform capitalism, where companies control significant parts of the economy through monopolistic infrastructure.

⁵² Gutta and Bhuyan, *Impact of Using Ride Hailing Apps on Social Connectedness between Auto Drivers*

⁵³ Elias and Gomez, *Digital Platforms as Institutional Actors in Global South*

⁵⁴ Sharma and Sharma, *The Gig Economy and Evolving Nature of Work in India*

⁵⁵ Joshi et al, *Informal and Shared Mobility: Status and Opportunities in India*

⁵⁶ Elias and Gomez, *Digital Platforms as Institutional Actors in Global South*

Gaps in Existing Research

Existing research has provided frameworks to understand transportation, social hierarchies, and platform labor in other major Indian cities. However, there is no comprehensive study that examines how rideshare platforms interact with the unique infrastructural, social, and labor conditions of Bengaluru specifically. First, infrastructure-focused research does not evaluate the ways rideshare platforms undermine Bengaluru's long-term transit planning goals. It does not evaluate whether it may divert riderships from Namma Metro or weaken TOD policies. Second, although there is provocative research on mobility inequalities in Bengaluru, it does not examine how rideshare platforms intensify these hierarchies. Third, while research on platform-based labor in India is extensive, none focus on how Bengaluru ride-hailing labor fits into the broader structure of platform capitalism in the city.

This research will understand how rideshare platforms reshape mobility, inequality, and labor in Bengaluru. It will ask the following questions:

1. How do rideshare platforms in Bengaluru undermine transportation infrastructural goals?
2. How do rideshare platforms in Bengaluru affect mobility inequalities? Which groups experience smoother transportation, and which groups suffer?
3. How does platform capitalism in Bengaluru through rideshare platforms affect driver livelihoods?

Methodology

To collect information on the impact of rideshare platforms, I traveled to Bengaluru, India under the University of Pennsylvania's Hassenfeld Grant for Research in Urban Studies. My extended family has lived in Bengaluru for nearly 80 years, with occupations such as medicine, IT, and teaching. I stayed with my 80 year-old grandmother for one month in August of 2025, and conducted many interviews along with her. I interviewed people of the following three backgrounds: 1) Drivers, 2) Passengers, and 3) Leaders in Transportation. Groups 2 and 3 were to understand how infrastructure and social behaviors have changed due to rideshare platforms, while with Group 1, I aimed to understand driver labor.

When interviewing drivers, I would call rides using one of the four major ride-hailing platforms in Bengaluru: Uber, Ola, Namma Yatri, and Rapido. The rides would be various lengths—some would be an hour, while some only ten minutes. The two possible modes of transportation are auto-rickshaws and four-wheeler cars; drivers from both modes were included in my analysis. I interviewed 15 drivers and conducted conversational interviews, noting information such as how long they have driven for, what their daily earnings are, experiences with safety and harassment, their opinions on the apps, whether they own their vehicle, and whom they support with their income. Every driver was very willing to participate and share information, and they were compensated 400 rupees for their time. The drivers spoke Kannada and Tamil, so I interviewed them in these languages, and translated them afterwards to English as I reviewed interview transcripts via NVivo.

I followed a similar, relatively informal format for passengers. Among these were three members of a family in my grandmother's neighborhood—a couple in their 50s, both professionals and long-term residents of the city, and their 83 year-old mother, whom I

interviewed at their home. Additionally, I interviewed four domestic workers that were employed in my grandmother's apartment complex, with occupations such as being a maid or cook. Finally, on the public bus, I approached people, in total asking around 15 men and women, with diverse income backgrounds, ages, and reasons for commuting.

Last, I interviewed field experts and leaders. I interviewed Dr. Ashish Verma, the founder of the Indian Institute of Science's Sustainable Transportation Lab, and Dr. Meghna Verma, the researcher behind a cutting-edge paper about how Bengaluru's free bus scheme for women impacts their safety. I also interviewed Mr. Srinivas Shenoy, the Head Warden of the Bengaluru Traffic Police. He oversees traffic crime management and wardens around the city. I interviewed Ms. Deepa Padmar, a labor rights lawyer at the Vidhi Centre for Legal Policy, who has been involved in cases for the Indian App-Based Workers Federation, helping create policy for the Karnataka Labor Department. To understand the perspective of platform leaders, I interviewed a Senior Data Scientist at Uber India. I also interviewed Mrs. Nagalaxmi, a former auto-driver-turned entrepreneur, who owns a startup that empowers women to become drivers of electric vehicle auto-rickshaws.

Data

From analyzing the interview transcripts via NVivo, the categories ‘physical infrastructure,’ ‘social hierarchies,’ and ‘driver labor’ became strong through multiple rounds of encodings.

From these codes, the following claims were yielded:

1. Existing efforts in traffic reduction in Bengaluru are car-centric and do not address broader infrastructural challenges, an attitude reflected in the mind of rideshare users.
2. Rideshare platform commuters and public transit users have vastly different travel experiences in terms of waiting times, reliability, and productivity.
3. Respondents working for rideshare platforms as drivers face hostile work environments: they encounter volatile wages, frequent deactivations, and a lack of support with harassment cases.

Permanent Infrastructural Changes

Rideshare platforms create pressures for drivers to break traffic safety rules, and they provide a superior alternative to public transit for its users. Meanwhile, traffic management efforts in Bengaluru prioritize policing human behavior.

“With all the people calling rides, there are more cars on the road, so there’s more congestion. It’s as simple as that,” a Data Scientist at Uber stated. In addition to the increase in vehicles due to ride-hailing services, the road is increasingly congested with food delivery platform employees. Many employees do not drive registered vehicles, and are actively rewarded for breaking traffic rules to complete their work on-time.

“Because of the time pressure and fear of bad ratings, drivers feel pressure to drive a bit recklessly—breaking signals, taking risky shortcuts, meandering through traffic. Since

they're often on two-wheelers, it's easy to maneuver. All these add up to congestion in Bangalore,” an Uber employee stated.

In efforts to understand how traffic was managed in Bengaluru, I reached out to the Bengaluru Traffic Police (BTP). During my meeting with the BTP, I visited their 5-story building, which had a museum in the basement about the history of the Bengaluru Traffic Police—a popular school field trip location. From the museum I learned that the BTP was created in 1930 under colonial rule, given the need to have a traffic-specific wing. The organization disciplines people for commuting traffic-related crimes, such as skipping red lights, not wearing a helmet or seatbelt, texting while driving, and speeding. They have complex technologies to monitor traffic and crimes—video cameras predicting traffic flow and changing signal colors accordingly, and facial recognition software to catch ‘traffic criminals.’ They have a three year-old program, “Catch them Young” where they teach children about “lane discipline.” They have simulation games teaching children how to be traffic wardens, and how to drive safely in the city. Aside from the museum, the top floor had large projection screens, with live camera footage of thirty busy intersections of Bengaluru. They were monitoring these intersections for safety purposes. A group of traffic wardens were constantly monitoring the camera footage. They had a Whatsapp chat with traffic wardens, to send them to the busy intersections in case of emergency. As I interviewed the Head Traffic Warden, I asked him about Bengaluru’s vision for traffic improvement. He emphasized their developed technology—their AI street cameras changing colors depending on vehicle flow, and ASTraM, standing for Actionable Intelligence for Sustainable Traffic Management, an app for traffic management. Their goal is to make ASTRaM a popular app with crowd-sourced information about traffic in Bengaluru, and it would provide a trip planning software more reflective of traffic. They also wanted residents to use

ASTRaM to report traffic criminals. The Head Traffic Warden stated that ultimately, infrastructure-related traffic issues was just “5% of the problem—most of it is because of lack of driver discipline. Nobody needs to drive on the wrong side of the road.” There are campaigns at busy intersections reminding drivers to “follow lane discipline,” as seen in Figure 3⁵⁷. However, the Head Traffic Warden stated that despite efforts, traffic has not decreased at all in recent years.



Figure 3. *Policing Efforts*. Photograph by the author, Bengaluru, August 2025

⁵⁷ Photograph taken by author during fieldwork in Bengaluru, August 2025

Aside from new efforts in traffic technology and policing, I interviewed rideshare platform users. Out of the five respondents who utilize rideshare platforms to get to work, none of them use public transport. Many major IT and finance companies provide carpool cab services for their employees. “My office isn’t near a metro stop anyways, so public transit wouldn’t be convenient. Most of us drive or use the company shuttle,” an employee of a company providing shuttles states. She praises the convenience and safety of these shuttles, as they use a digital app to organize the drop-off points, and be sure to drop off women first. “If I’m not using my office-provided cab, I call an Uber,” she states. Another IT employee states, “I use Uber every single day to get to work.” When asked about public transportation, he stated the following:

“It takes the same amount of time to commute with the Metro or to take an Uber, so why would I suffer on the Metro for no reason? I’d much rather have the driver come to my door, work in the car, and get dropped– than be in a crowded space, and then walk a mile to get to the office.”

When considering the impacts of rideshare platforms on Bengaluru infrastructure given the interview data, we can understand that they create stressful driving conditions for drivers, they are not considered in citywide anti-traffic measures, and that they are a daily solution viewed as superior to public transit by its users.

Inaccessibility of Transportation

Meanwhile, the Bengaluru public bus is crowded and gender-segregated. The city-wide Shakti scheme allows all female residents of Bengaluru to ride on the public bus free of charge. The front half of the bus is reserved only for women, while the back half is for men, with a

preference for women if the first half is full, as seen in Figure 4⁵⁸. This goes beyond a social norm—there are labels in the bus, as seen in Figure 5⁵⁹.

Bus users have experiences very different from rideshare users. Most of the time, the bus stations are not sheltered, as seen in Figure 6⁶⁰. Two women had been sitting at the curb for three hours, waiting for the bus to come. They both worked in domestic service as maids, and always traveled to and from work together, to protect one another. They said they were used to waiting hours for the bus, as that was a part of their daily routine. Another woman working in domestic service stated that she normally would cycle around two kilometers to work on her personal bike, but she took the bus due to weather conditions that particular day.



Figure 4. 5 PM at the BMTC Bus in Old Bengaluru with Labeled Gender Sections.

Photograph by the author, Bengaluru, August 2025

⁵⁸ Photograph taken by author during fieldwork in Bengaluru, August 2025

⁵⁹ Photograph taken by author during fieldwork in Bengaluru, August 2025

⁶⁰ Photograph taken by author during fieldwork in Bengaluru, August 2025



Figure 5. *Label of Ladies Section on BMTC Bus.* Photograph by author, Bengaluru, August 2025



Figure 6. *Waiting for the bus.* Photograph by author, Bengaluru, August 2025

It is difficult to get a seat on the public bus, but a community-minded culture exists—people always gave up their seats for senior citizens and women. One 80 year-old resident, who has resided in Bengaluru for over fifty years stated that she would take the bus when she was younger. She praised its social environment, stating “on the bus, I always have friends. Sometimes I will make friends on the bus, in the ladies’ section.” She no longer takes the bus due to her old age, and utilizes rideshare-hailed auto-rickshaws when she commutes one time per week. She commented on how the social nature of auto-rickshaws have changed for her. She disliked that apps required her to input her destination address, as she wouldn’t always know. “Back in the day, I would just tell the auto driver, go right, go left. There was no formal address and all,” she stated.

In addition to auto-rickshaw navigation culture shifting, the culture of personal drivers has become obsolete. In the past, affluent families would purchase vehicles and hire a driver to drive them places, on-demand. The driver would be paid weekly, and would typically wait in the vehicle for their passenger. “Our driver used to be like another family member to us—we would chat every day when I’d go to work, and he would treat my daughter like his own. But we realized it was cheaper to just call Ubers every time—so we let him go,” an affluent, long-term resident stated.

The transportation experience of rideshare users is of stark difference to those using the public bus. Rather than spending hours waiting on a curb for the bus, a daily Uber user stated, “The driver comes all the way to my gate, so I don’t have to go on the dirty road anymore. I will call the ride and then get ready at home. App-based autos are definitely more expensive than what you’d hail on the street, but that’s just the cost you pay for comfort.”

One respondent mentioned that she does not mind paying an extra fee if it means improved safety. “Sure, it’s a cost, but I can send the live route of my ride to my husband. He is always aware of where the cab is taking me,” a middle-aged, affluent woman stated.

When asked about the intended demographic of rideshare platforms, a Data Scientist at Uber stated the following:

“Our target audience is definitely not the mass market because the fares per kilometer are quite high. It’s predominantly catering to a more affluent segment where affordability isn’t a huge problem. There’s a large proportion of people who can’t afford ride-hailing services. They depend on public transport. Rideshare platforms are designed for the comfort of those who can afford it.”

Driver Labor Precarity

Rideshare platforms have entirely changed the labor experience for drivers. Out of the 15 drivers interviewed, the older auto-rickshaw drivers formerly found passengers only through getting hailed on the street. They would use a meter that would run during the ride, as pictured in Figure 7⁶¹. Most drivers have the meters covered in plastic as they are no longer used given rideshare apps determining the price.

Out of the 15 drivers interviewed, 10 had been driving prior to the rideshare apps entering the transportation system. They did not own smartphones prior, but two of them got a free smartphone from Uber in exchange for signing up to be a driver. When initially asked about their opinions of the platforms, many drivers stated, “without the apps, there would not be enough passengers.”

⁶¹ Photograph taken by author during fieldwork in Bengaluru, August 2025



Figure 7. *Back of an Auto-Rickshaw*. Photograph by author, Bengaluru, August 2025

Every driver interviewed had four ride-hailing apps downloaded that they would switch between: Uber, Ola, Rapido, and Namma Yatri. They mentioned Uber having more passengers than the other platforms, but paying slightly less per ride, yet driver earnings were unpredictable regardless of the specific platform utilized. When asked about driver commission algorithms, a Data Scientist at Uber stated, “if we don’t keep the commission according to the demand, drivers will become lazy.” The driver commissions are based on the demand from passengers—a five kilometer ride would pay a driver differently at 5 PM versus 7 AM.

Drivers bring in substantial revenue after working 12-14 hours per day, but a large amount gets taken out for commissions, platform rent, and rising fuel costs. “Commission each day is different. Money is taken accordingly without a strict percentage,” a driver stated. “My daily revenue is around 4000 rupees. It ends up at around 2000 with all the costs,” another driver stated. One driver showed me the pricing scheme of Rapido, where there were different ‘tiers’ of memberships, priced with higher levels if a driver wanted more rides allocated. Platforms

explicitly describe their pricing schemes as ‘trade secrets.’ Mr. Javid, an auto-rickshaw platform worker and the General Secretary of the Bengaluru Auto-Rickshaw Workers’ Union, states that ‘if the drivers don’t pay their commission, their account will get deleted. It is the app’s discretion to cancel a driver from the platform if they deviate from routes too often.” He stated one of the Union’s main grievances is the frequent banning of drivers from platforms, without prior warning or discussion. Every driver interviewed felt upset at the low commission rates, and recognized that they needed to simply work more hours in order to keep up profits, as fuel costs increase by the day. A driver I interviewed had the poster in Figure 8 in the backseat of his car⁶². He stated he would prefer if people called him when they needed a ride, rather than hailing through the app so he could earn more profits.



Figure 8. *Bengaluru is Anti-Pricing Algorithms*. Photograph by author, Bengaluru,

August 2025

⁶² Photograph taken by author during fieldwork in Bengaluru, August 2025

Pricing algorithms are frustrating for both drivers and passengers. Compared to other local Indian platforms, Uber charges less for each ride, perhaps due to its larger-scaled operations. Figure 9 shows the same ride, being hailed via Uber and Ola. The price from Uber is 20% less than that of Ola's⁶³.

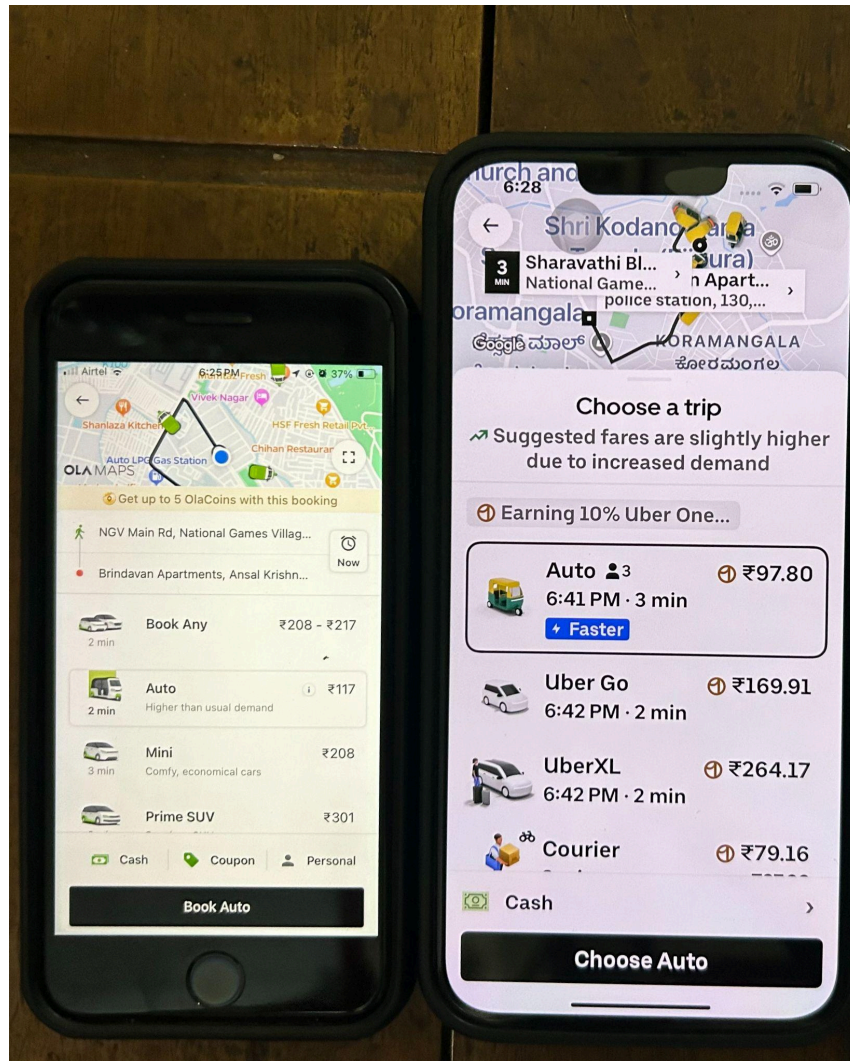


Figure 9. *Same ride, different apps*. Photograph by author, Bengaluru, August 2025

⁶³ Photograph taken by author during fieldwork in Bengaluru, August 2025

In addition, as seen in Figure 10, Uber has an option when hailing rides to add an additional amount to the existing fare to increase chances of getting a ride, marketing this as going “straight to the driver”⁶⁴

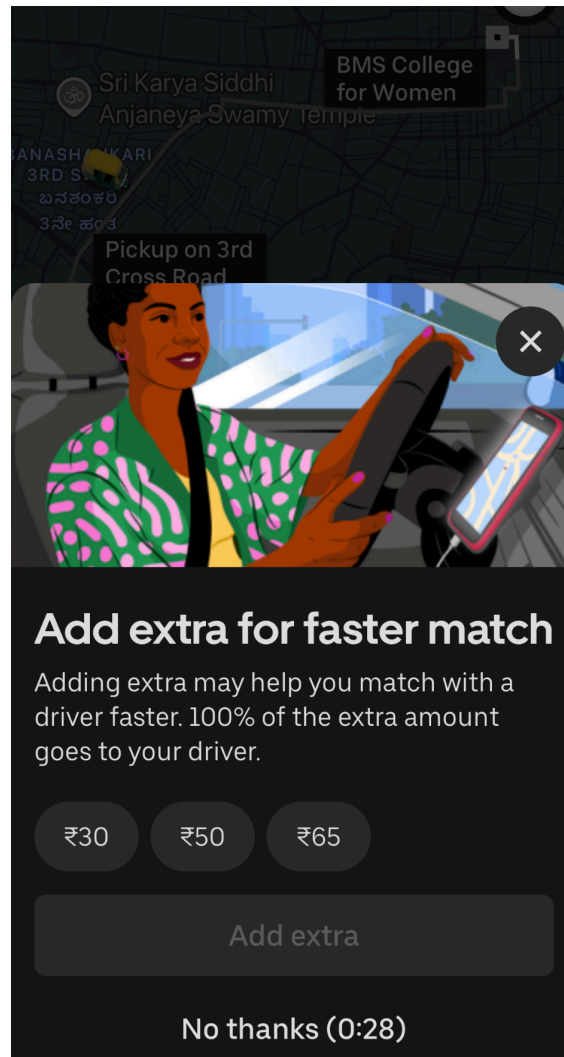


Figure 10. *An Algorithmic Distraction*. Photograph by author, Bengaluru, August 2025

Aside from volatile pricing and commission algorithms, drivers also experience violence from passengers and unsafe work environments. Out of the 15 drivers interviewed, they all had a response when asked ‘have you ever felt unsafe with a passenger?’ One driver reported having

⁶⁴ Photograph taken by author during fieldwork in Bengaluru, August 2025

his phone and wallet stolen by a passenger, and being threatened with a knife. Many stated that their fellow drivers have all reported having their phones or cash stolen by passengers. “Apps said they will not take responsibility for safety issues. If I want to get any money back, I go to the police,” a driver stated. Many reported Uber’s platform being difficult to navigate when reporting safety issues with passengers, and they would often get redirected without any results. When asked about driver safety protections to an Uber Data Scientist, he emphasized that they “had systems in place,” but did not provide any additional concrete information when asked.

In addition to passenger unsafety, work environments for drivers are one of the most hazardous in Bengaluru. Drivers reported problems of not having proper public restrooms. “Public parks are usually closed midday for lunch. So where can I rest? I just sit in the auto, and wave away people asking for a ride,” a driver reported. Mr. Javid stated the following about health:

“Looking at the app while driving is just not safe. You have to keep looking down at the phone, and directions will change because passengers will cancel rides. It hurts the eyes and brain. Drivers are never healthy people. We try to give insurance to our drivers, but it’s hard.”

Every driver interviewed thus far was a male—one even reported that there was “no chance” that a woman could become a driver, given its unsafe nature. There are 100 female auto-rickshaw drivers in Bengaluru, and two were interviewed. Namma Yatri, a rideshare platform local to Bengaluru that does not take commissions, has an offshoot company called Montra that employs women to drive easy-to-manuever electric auto-rickshaws. Through training on how to drive and how to protect oneself from dangerous situations, the Montra employees feel empowered by their occupation. When interviewed, they reported that all of their

fellow Montra drivers face problems at home, such as their husbands being ill, passed, or unable to work, and needing a greater source of income to support their families. Many women were formerly garment workers or worked in domestic cleaning services, and prefer driving over these occupations where they are easily harassed and exploited. “At least driving is done in public—I don’t have to go into someone’s home to clean anymore,” a driver said.

Discussion and Analysis

According to the interviews, it is clear that rideshare platforms have created significant changes in Bengaluru's landscape: they have contributed to car-centric, bottom-up infrastructure in the city, they have made a larger gap in transportation experiences by class, and they have changed the labor experience of drivers.

Physical Infrastructure

From the interviews with traffic management leaders, the Bengaluru Traffic Police efforts are very bottom-up, prioritizing policing and new technology developments rather than addressing root problems causing traffic. Billboard signs and schoolchildren campaigns to 'follow lane discipline' place the blame for traffic congestion on drivers, rather than root infrastructural issues. New technologies like crowd-sourced apps to determine traffic hotspots, AI traffic signals, and simulations for schoolchildren to learn how to be traffic wardens are heavily funded. Meanwhile, potential root issues such as rideshare platforms encouraging reckless driving, or the refusal of car-users to switch to public transit, are not considered. The lack-of-driver-discipline sentiment diverts attention from infrastructural and algorithmic concerns that encourage such behavior. This illustrates the elitist, car-centric approach to transportation management that existing literature mentioned. By fining a rideshare driver 500 rupees for making an illegal turn during rush hour, the most vulnerable are punished, rather than those in power taking accountability. Rideshare platforms undoubtedly cause traffic congestion, given that the business model is based upon quick, efficient rides, and that many vehicles are unregistered, encouraging them to break traffic rules. Rideshare platforms and regulation must be considered in traffic management efforts.

Many developing cities, especially those in India were characterized by their ‘informal’ modes of transportation that involved negotiation, carrying more people than their capacity, and being unregistered. This formerly complemented Bengaluru public transportation when public buses could not carry the entire population. However, this infrastructural landscape has entirely changed in recent times: the existing systems can no longer efficiently mobilize the population, a rapid metro system is currently being implemented, and these ‘informal’ systems have now become algorithmically-controlled. This recent change in the infrastructure has not been taken into account in city-wide planning efforts. Rideshare platforms will be a hindrance to full mass public transit implementation in Bengaluru. Rideshare users are currently very apprehensive about public transit, and will not switch to public transit unless first and last-mile connectivity issues were solved, and there was a significant time difference between the two modes. While cost is not a concern for most users, the time savings are important. The City of Bengaluru’s stated transportation goals involve increasing Metro ridership, but it does not consider how existing modes of rideshare will impact Metro ridership. The convenience of rideshare apps and its marginal cost to affluent users will deter a large segment of the population from using public transit, which may present challenges as the Metro continues to develop.

Ultimately, the interviews proved that rideshare platforms do create infrastructural changes by deterring public transit use and contributing to traffic congestion, yet existing traffic management projects do not consider their impacts.

Social Infrastructure

Bengaluru is being pushed towards car-centric, digital platforms for daily commutes. This means two things: driver work is exploited, and commuting experiences are increasingly class-segregated.

Driver-passenger social relationships have changed. In the interviews, affluent respondents mentioned no longer having personal drivers, and instead only using rideshare platforms for daily commutes. Prior to rideshare apps, drivers would interact with passengers—having humorous fair negotiations, discussing directions, and complaining about traffic. With rideshare platforms, riders often get into vehicles without speaking a word to drivers. Drivers now travel to all parts of the city, according to where the app instructs them to go, rather than being local to a neighborhood and developing social ties with the community. Passengers feel empowered to cancel rides if they get a better fare on a different platform, as it is free to cancel rides and they feel no obligation to the drivers. This formerly personable, social relationship has become disposable.

In addition to the lack of social passenger interactions, drivers feel vulnerable with safety. The cars have no physical safety feature: no emergency buttons, or no consistent number to call in case of an emergency. The drivers noted getting no protection from rideshare platforms in the event of emergencies, as they would keep redirecting them and asking them too many questions—prolonging the process of compensation. Although the drivers work for up to 16 hours on the platforms, they are not offered health insurance or sick leave protections.

Additionally, the pricing scheme of the platforms is unclear to both drivers and passengers, leaving both parties frustrated. Uber, given its standing as the only global rideshare platform, is likely able to price their rides as less compared to other platforms given their

monopoly on rideshare trips. They are able to pay drivers less as they offer the most number of rides to drivers due to their wider market reach. Drivers have no choice but to use their platform in order to get rides. They have complex pricing and matching algorithms, yet distract users from this by asking for tips given directly to drivers to increase their chances of getting a match. As seen in the advertisement for a better rideshare platform with no complex algorithm, both drivers and passengers want transparent algorithms. Surge pricing makes passengers pay more, while drivers get paid relatively similar amounts to drive in conditions with more traffic. Nearly every driver interviewed was unsure of their exact commission or project earnings per day—a sign of opaque algorithms and exploitations.

The problem of opaque algorithms and workers on wheels extends beyond rideshare platforms in Bengaluru—home delivery platforms are in abundance, with thousands of disposable, unregistered drivers racing around the city in two-wheelers, pressured by app metrics. Apps such as Swiggy, Dunzo, and Blinkit are similarly of little cost to affluent users, but allow for unsafe driving and unstable earnings.

Aside from platform labor in Bengaluru, this is part of a greater global concern of ‘platform capitalism,’ where digital platforms act as intermediaries in everyday economic, social, and labor transactions⁶⁵. Although these platforms find support among some scholars that equate their adoption with a utopian vision of efficiency, peer-to-peer participation, and democracy, many scholars also note the much more serious concentration of social, political, and economic power, labor precarity, and surveillance that such platforms bring with them⁶⁶. Other online contract work sites such as AirBnb and Facebook Ads are criticized as monopolistic and unfair to the workers that use them – such companies keep disproportionate amounts of the revenue that

⁶⁵ Casas-Cortés, *Platform Capitalism*

⁶⁶ Casas-Cortés, *Platform Capitalism*

workers bring in, simply for running the platform. As Uber has disrupted long-term transportation ecosystems in Bengaluru, AirBnb has destabilized housing markets elsewhere. In Barcelona, residents led mass protests against AirBnb for gentrifying neighborhoods and displacing local families, painting graffiti against the company throughout the city⁶⁷. AirBnb and Uber have both also shifted upfront cost risks to everyday workers that now must purchase their own cars and real estate to earn their livings on these apps, where two decades ago such costs may have been borne by such companies. Nowadays, these platform providers simply take a large sum from transactions for facilitating a customer-provider relationship. Although vehicle ownership may have reduced amongst affluent residents, the burden of buying vehicles is now placed on drivers, facing vulnerabilities given such large upfront costs.

In addition to exploiting driver labor, there is a wider gap of transportation experiences between commuters of different socioeconomic backgrounds. Low-income women interviewed waited for the bus by sitting on the curb for hours and cycled in extreme heat to get to work, while high-income men received rideshare, air-conditioned vehicles waiting at their gate. Commutes that high-income residents would conduct by auto-rickshaws are done by walking or biking for low-income residents—also creating more vehicle congestion. It is in the best interest of all residents to shift towards more sustainable means of transportation, as even high-income residents spend hours in their cars commuting short distances. As discussed in the literature review, a long-term solution would involve mass public transit of all residents. Rideshare platforms discourage affluent residents from using public transportation. In order to shift rideshare users towards public transportation, their use must be disincentivized by making them significantly more expensive, in tandem with a serious investment in fast public transit with a focus on last-mile connectivity. The existing literature suggested that informal transportation, in

⁶⁷ Martinez, *Citizens, Squatters, Homeowners and Tenants: Democratization and Capitalism in Spain*

some cases, motivates public transit to improve its conditions, in a competitive nature. However, in Bengaluru, this is not the case: it has created two entirely separate ‘classes’ of transportation. By having a streamlined mode of transportation across classes, there is a greater chance of achieving equal opportunities amongst all residents.

Ultimately, the private ownership of transportation does not create a sustainable, long-term solution. When transportation modes prioritize profits, they do not serve low-income residents, and they are left to take less efficient modes. Aside from creating a new aspect to social inequalities, profit prioritization also leads to roads being filled to the brim with low-occupancy vehicles and customers seeking unsustainable, luxurious transit journeys. Given the broad unpopularity of traffic congestion – it is an issue that affects all social and economic strata – understanding it as an issue within ever-changing capitalist profiteering practices may illustrate its true nature to more Bengaluru residents.

Conclusion and Future Research

Rideshare platforms have an inherently negative impact on Bengaluru in three manners: first, by contributing to car-centric, traffic-congested infrastructure, second, by creating a wider gap in transportation experiences by class, and third, by exploiting driver labor.

As Bengaluru develops a rapid metro system, the impact of rideshare platforms on mass transit ridership must be considered. Many users of rideshare platforms in Bengaluru are very apprehensive about converting to public transportation, suggesting that the ease of access to these platforms has made Bengaluru more car-centric. The role of rideshare platforms has not been addressed in any transportation plans, but given their prominence and far-reaching impacts in physical and social infrastructure, they must be considered.

Conclusions of this paper extend beyond Bengaluru social infrastructure. Digital platforms such as Facebook Marketplace, AirBnb, Uber, and Ebay are making urban life across the world more precarious and unlivable. As technology continues to develop unrestrained, its acceleration of class divides, exploitation, and social inequalities must be particularly scrutinized.

Additionally, when conducting research, I initially intended to understand the impact of rideshare apps on women's safety, but it was surprising to find that women respondents did not share any discomfort while in rideshare vehicles or on public transit. Perhaps there has been more progress on this front in Bengaluru. However, studies conducted in other Indian cities suggest this is still a systemic problem; I believe my lack of data here may have been due to limitations on the amount of time I spent with my interviewees.

Future studies could determine how Transit-Oriented Development frameworks could be adopted in Bengaluru, given its existing history with car-centric development. Alternatives to

TOD are also valuable; a stronger framework of sustainable transportation for people of all economic backgrounds in a city that is developed and has a high existing population density, needs to be researched. There is yet to be a widely accepted transit development framework to be used in populous South Asian cities relying on public buses and informal transportation. Additionally, as the metro system continues to develop, further research should seek to specifically understand how to shift trips away from private vehicles and onto public options.

Transit issues in Bengaluru will not be easy to solve, with its class stratification, income inequalities, motor vehicle-centered infrastructure, and population of 14 million people. A sustainable transportation system must prioritize mobilizing people of all economic backgrounds, rather than offering luxurious and affordable options for affluent people, while the majority of its residents are second-class commuters.

Bibliography

- Baas, Michiel. "Bangalore and Its IT Industry: A Changing Landscape of Work and Life in Urban India." *Economic and Political Weekly*, vol. 52, no. 24, 2017, pp. 31–37. *JSTOR*, <http://www.jstor.org/stable/26695992>. Accessed 25 Nov. 2025.
- Barry, Ellen. "In Case That Transfixed a Nation, Court in India Convicts Four Men of Rape." *New York Times*, 2013.
<https://www.nytimes.com/2013/09/11/world/asia/four-men-convicted-in-rape-case-that-transfixed-india.html>.
- Best, Richard A. "Implementing a Multimodal Vision: An Update of Arterial Transportation Management in Arlington County, Virginia ." <https://www.bluemontcivic.org/>. Accessed September 27, 2025. <http://www.bluemontcivic.org/docs/CB07C2001.pdf>.
- Cervero, Robert B. "Linking Urban Transport and Land Use in Developing Countries." *Journal of Transport and Land Use*, vol. 6, no. 1, 2013, pp. 7-24.
- Choubey, Shantanu Braj. "The Uber Conundrum: Analysing the Worker Rights of Uber Drivers in India." *National Law School Journal* 17, no. 1 (2023): 49–65.
<https://doi.org/10.55496/hvrg7750>.
- "Fairwork India Ratings 2020: Labour Standards in the Platform Economy." Fairwork, 2020.
https://fair.work/wp-content/uploads/sites/17/2021/01/Fairwork_India_2020_report.pdf.
- Fileborn, Bianca, Elena Cama, and Alison Young. "Perceptions of Safety among Taxi and Rideshare Service Patrons: Gender, Safekeeping and Responsibilisation." *International*

Journal for Crime, Justice and Social Democracy 11, no. 1 (April 20, 2022).

<https://doi.org/10.5204/ijcjsd.2085>.

Frank, Andre Gunder. *The Development of Underdevelopment*, 1966.

Gopakumar, Govind (director). *A Social Life of a Bus*. Documentary film, produced by Govind Gopakumar, 2015

Gopakumar, Govind. *Installing Automobility: Emerging Politics of Mobility and Streets in Indian Cities*. MIT Press, 2020.

Giridharadas, Anand. *Winners take all: The elite charade of changing the world*. London: Penguin Books, 2020.

Gupta, Vipin & Qiu, Renfeng. (2013). The Rise of the Indian Multinational Corporations and the Development of Firm-Specific Capabilities. *Journal of Business Theory and Practice*. 1. 45. 10.22158/jbtp.v1n1p45.

“Indian Federation of App-Based Transport Workers Letter to Labour Department, Government of Karnataka.” Vidhi Centre for Legal Policy, August 8, 2024.

<https://vidhilegalpolicy.in/research/comments-on-the-draft-karnataka-platform-based-gig-workers-bill-2024/>.

Joseph B, Joseph M. Rise of the 'Gig Economy' and its Health Toll on Workers. *Indian J Occup Environ Med*. 2025 Apr-Jun;29(2):85-86. doi: 10.4103/ijoem.ijoem_217_25. Epub 2025 Jul 1. PMID: 40761717; PMCID: PMC12318557.

Knowles, Richard D., Fiona Ferbrache, and Alexandros Nikitas. "Transport's Historical, Contemporary and Future Role in Shaping Urban Development: Re-evaluating Transit Oriented Development." *Cities*, vol. 99, 2020, article 102607.

Kumar, Mayank. "Connected but Unprotected: Reimagining Decent Work for India's Urban Gig Workers." Pravasi Setu Foundation, August 11, 2025.
<https://psfresearch.com/connected-but-unprotected-reimagining-decent-work-for-indias-urban-gig-workers/>.

Manjoo, Farhad. Uber wants to rule the world. First it must conquer India. - The New York Times, April 14, 2017.
https://www.nytimes.com/2017/04/14/technology/uber-india.html?_r=0&mcubz=0.

Mchunu, Gugu G., Desmond Kuupiel, Busisiwe P. Ncama, Christopher Isike, Marcel Kistan, Julian David Pillay, and Sinegugu E. Duma. "Public Transport Systems and Safety of Female Commuters in Low-and-Middle-Income Countries: A Systematic Scoping Review." *BMC Women's Health* 25, no. 1 (May 29, 2025).
<https://doi.org/10.1186/s12905-025-03821-0>.

Muniraja N. and R. Thimmarayappa. "A Study on Social and Economic Benefits of Bengaluru Metro Rail Transportation." *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 10, no. 11, Nov. 2023, www.jetir.org/papers/JETIR2311152.pdf.

Prakash, KM, Manoj RS, M Ramesh, Megha Patri, and Latha M. "Women Safety Cab." *International Research Journal of Modernization in Engineering Technology and Science*, October 6, 2023. <https://doi.org/10.56726/irjmets45094>.

Simet, Lena. “The Gig Trap.” Human Rights Watch, June 24, 2025.

<https://www.hrw.org/report/2025/05/12/the-gig-trap/algorithmic-wage-and-labor-exploitation-in-platform-work-in-the-us>.

Thondoo, Meelan & Márquez, Sandra & Nieuwenhuijsen, M & Marquet, Oriol. (2020). Small cities, big needs: Urban transport planning in cities of developing countries. *Journal of Transport & Health*. 19. 100944. 10.1016/j.jth.2020.100944.

Verma, Ashish, S. Sreenivasulu, and N. Dash. “Achieving Sustainable Transportation System for Indian Cities – Problems and Issues.” *Current Science* 100, no. 9 (2011): 1328–39.

<http://www.jstor.org/stable/24076598>.

Verma, Meghna, M. Manoj, Nikhita Rodeja, and Ashish Verma. “Service Gap Analysis of Public Buses in Bangalore with Respect to Women Safety.” *Transportation Research Procedia* 25 (2017): 4322–29. <https://doi.org/10.1016/j.trpro.2017.05.283>.

Verma, Meghna, Nikhita Rodeja, M. Manoj, and Ashish Verma. “Young Women’s Perception of Safety in Public Buses: A Study of Two Indian Cities (Ahmedabad and Bangalore).” *Transportation Research Procedia* 48 (2020): 3254–63.
<https://doi.org/10.1016/j.trpro.2020.08.151>.